CS & DA

Python-For Data Science

Python Datatypes

DPP : 02

Q1 The output of below Python Code Segment is _____

```
a,b=1,3
for i in range(a,b):
    for j in range(b,a,-1):
        print(i+j, end=' ')
```

(A) 4,3,5 (B) 4,5,3,4
(C) 4,3,5,4 (D) 4,5,3

Q2 The final value of result is _____

```
result=1
for x in range(result,4,2):
    result+=1
    for y in range(result):
        result+=2
print(result)
```

Q3 What is printed by below Python Code Segment?

```
for i in range(5,2,-1):
    i=10
    for j in range(1,3):
        i=i+j
print(i)
```

(A) 12 (B) 13
(C) 15 (D) 16

Q4 What will be final ans value from the below code?

```
a=1
b=4
ans=1
while a<4:
    ans=ans+a
    while b>1:
        if b<=2:
            break
        ans=ans+b+a
        b=b-1
    a=a+2
print(ans)
```

Q5 The total number of times print statement is executed is _____

```
for i in range(7,8):
    for j in range(6,7):
        print(i*j)
```

(A) 1 (B) 2
(C) 3 (D) 4

Q6 The output of below code is ____

```
i=print('GATE',end='')
while i:
    print(i,end=',')
    i=i-1
```

(A) Error (B) GATE 4,3,2,1
(C) GATE 4,3,2 (D) GATE

Q7 What will be the output of below Python Code Segment?

```
for i in range(1,10,2):
    if i%3 ==0:
        print(i+2,end=' ')
        continue
    elif i%5==0:
        print(i-1,end=' ')
        break
    else:
        print(i+1,end=' ')
```

(A) 2, 5 (B) 1, 4, 3
(C) 2, 5, 4 (D) 2, 5

Q8 The value of 'i' printed by below code is _____--

```
i=5
for j in range(5):
    match j:
        case 2: i=i+2
        case 4: i=i-1
        case 5: i=i+1
```

case 3: i=i-2
case 1: i=i+i

print(i)

Answer Key

Q1	C
Q2	21
Q3	B
Q4	14
Q5	A
Q6	D
Q7	C
Q8	9

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

1. **Variable Initialization**:
   - a = 1
   - b = 3
2. **Outer Loop**:
   - for i in range(a, b) translates to for i in range(1, 3). So, i will take the values 1 and 2.
3. **Inner Loop**:
   - for j in range(b, a, -1) translates to for j in range(3, 1, -1). So, j will take the values 3 and 2.
4. **Print Statement**:
   - print(i + j, end=' ') prints the sum of i and j with a space as the separator.

Iteration Details:

- **When i = 1**:
  - j = 3: print(1 + 3) => 4
  - j = 2: print(1 + 2) => 3
- **When i = 2**:
  - j = 3: print(2 + 3) => 5
  - j = 2: print(2 + 2) => 4

Output Generation:

Combining all the prints from the loops:

- For i = 1: Print 4 3
- For i = 2: Print 5 4

So, the complete output is: 4 3 5 4

Conclusion:

The correct answer is:
(C) 4 3 5 4

Q2 Text Solution:

Initial Values:

- result = 1

Outer Loop Analysis:

1. **Outer Loop for x in range(result, 4, 2)**:
   - The range function will generate values starting from result, up to but not including 4, with a step of 2.
2. **Iteration 1**:
   - result = 1
   - The outer loop range(1, 4, 2) generates 1 and 3 (i.e., x will take these values).

Sub-Iteration 1 (for x = 1):

- result += 1 result becomes 2
- Inner loop: for y in range(result) translates to for y in range(2) (i.e., y takes values 0 and 1).
  - **Inner Loop Iteration 1 (for y = 0)**:
    - result += 2 result becomes 4
  - **Inner Loop Iteration 2 (for y = 1)**:
    - result += 2 result becomes 21

Q3 Text Solution:

1. **Outer Loop (for i in range(5, 2, -1))**:
   - This loop generates values starting from 5 down to 3 (inclusive). However, the values of i in the outer loop are immediately overwritten by i = 10 inside the loop.
2. **Inside the Outer Loop**:
   - i is set to 10 at the start of each iteration.
   - **Inner Loop (for j in range(1, 3))**:

- This loop generates values 1 and 2.

3. **Updates within the Inner Loop**:
   - On each iteration of the inner loop, i is updated by adding the value of j.

Detailed Execution:

For each iteration of the outer loop:

- **Iteration 1 (for i = 5)**:
  - i = 10 (as set inside the outer loop)
  - **Inner Loop**:
    - j = 1: i = 10 + 1 = 11
    - j = 2: i = 11 + 2 = 13
- **Iteration 2 (for i = 4)**:
  - i = 10 (as set inside the outer loop)
  - **Inner Loop**:
    - j = 1: i = 10 + 1 = 11
    - j = 2: i = 11 + 2 = 13
- **Iteration 3 (for i = 3)**:
  - i = 10 (as set inside the outer loop)
  - **Inner Loop**:
    - j = 1: i = 10 + 1 = 11
    - j = 2: i = 11 + 2 = 13

After all iterations of both loops, the final value of i is 13.

The code prints 13.

Q4 Text Solution:

Initial Values:

- a = 1
- b = 4
- ans = 1

Execution Breakdown:

1. **Outer while loop**: while a < 4
   - Initially a = 1, so this loop will run as long as a is less than 4.
2. **First Iteration of Outer while loop (a = 1)**:
   - ans = ans + a ans = 1 + 1 = 2
   - **Inner while loop**: while b > 1
     - Initially b = 4.
     - **First iteration of Inner while loop (b = 4)**:
       - if b <= 2 is False, so it proceeds.
       - ans = ans + b + a ans = 2 + 4 + 1 = 7
       - b = b - 1 b = 3
     - **Second iteration of Inner while loop (b = 3)**:
       - if b <= 2 is False, so it proceeds.
       - ans = ans + b + a ans = 7 + 3 + 1 = 11
       - b = b - 1 b = 2
     - **Third iteration of Inner while loop (b = 2)**:
       - if b <= 2 is True, so it breaks out of the inner loop.
   - a = a + 2 a = 1 + 2 = 3
3. **Second Iteration of Outer while loop (a = 3)**:
   - ans = ans + a ans = 11 + 3 = 14
   - **Inner while loop**: while b > 1
     - b is now 2 (from the previous iteration), so b <= 2 immediately results in breaking out of the inner loop.
   - a = a + 2 a = 3 + 2 = 5
4. **Termination**:
   - a = 5, which is no longer less than 4, so the outer while loop terminates.

Final Value of ans:

After the loops complete, the final value of ans is 14.

The final value of ans is 14.

Q5 Text Solution:

1. **Outer Loop (for i in range(7, 8))**:
   - The range(7, 8) generates only one value: 7. Therefore, the outer loop will iterate once with i = 7.
2. **Inner Loop (for j in range(6, 7))**:
   - The range(6, 7) generates only one value: 6. Therefore, the inner loop will also iterate once with j = 6.

Execution:

- For i = 7 and j = 6, the print statement is executed once, producing the result of 7 * 6.

Conclusion:

Since the inner loop executes once for each iteration of the outer loop and the outer loop also executes once, the print statement is executed exactly **1** time.
So, the correct answer is:
(A) 1

Q6 Text Solution:

1. **First Statement**:
   - i = print('GATE', end='')
   - The print function outputs GATE without a newline (due to end=''), and the result of print() is None. Therefore, i is assigned the value None.
2. **while i Loop**:
   - The while loop condition is while i, which will evaluate to True if i is not None (or any non-zero/non-null value). Since i is None, the condition while i evaluates to False.
3. **Execution**:
   - Since i is None and the loop condition while i is False, the while loop will not execute.

Output:

- The only output from this code is GATE from the print('GATE', end='') statement.
- No additional output is generated because the while loop does not execute.

Conclusion:

The output of the code is simply:
(D) GATE

Q7 Text Solution:

1. **Loop Initialization**:
   - The for loop iterates with i taking values from the range(1, 10, 2), which generates 1, 3, 5, 7, 9 (odd numbers from 1 to 9).
2. **First Iteration (i = 1)**:
   - i % 3 is 1 % 3 = 1 (not 0), so the if condition is not met.
   - i % 5 is 1 % 5 = 1 (not 0), so the elif condition is not met.
   - The else block executes: print(i + 1, end=' ') print(1 + 1, end=' ') prints 2.
3. **Second Iteration (i = 3)**:
   - i % 3 is 3 % 3 = 0, so the if condition is met.
   - print(i + 2, end=' ') print(3 + 2, end=' ') prints 5.
   - continue statement skips the remaining code and moves to the next iteration.
4. **Third Iteration (i = 5)**:
   - Since the continue statement was executed in the previous iteration, this iteration is not reached. The loop ends before this point.

Conclusion:

The output of the code is:
2 5
So, the correct answer is:

(D) 2, 5

Q8 Text Solution:

Breakdown:

1. **Initial Value**:
   - i = 5
2. **Loop through j in range(5)**:
   - range(5) generates values 0, 1, 2, 3, 4.
3. **Match Cases**:
   - **For j = 0**:
     - No case matches 0, so i remains 5.
   - **For j = 1**:
     - case 1: i = i + i i = 5 + 5 = 10.
   - **For j = 2**:
     - case 2: i = i + 2 i = 10 + 2 = 12.
   - **For j = 3**:
     - case 3: i = i - 2 i = 12 - 2 = 10.
   - **For j = 4**:
     - case 4: i = i - 1 i = 10 - 1 = 9.
4. **Final Value of i**:
   - After the loop completes, the value of i is 9.

Conclusion:

The value of i printed by the code is 9

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